Mingshi Chen

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EDUCATION

Rice University Houston, TX

M.S. in Electrical and Computer Engineering Aug 2024 - Dec 2025

Fudan University Shanghai, China

B.S. in Electronic Information Science and Technology Sep 2020 - Jun 2024

RESEARCH PROJECTS

DuoGate: Action and Context Control for Reliable Language Agents

Researcher | San Diego, CA | Jan 2026 - Present

Manuscript submitted to ICLR 2027; under review.

- Developed DuoGate, a framework coupling evidence-based action control with selective context adaptation for tool-using language agents. Connected execution outcomes and intervention explanations to reflection and curation, allowing corrected interactions to inform reusable guidance.
- Implemented checks for argument grounding, authorization, outcome support, and completion evidence, with recovery directed toward detected gaps. Designed separate gates for learning eligibility and playbook admission to determine when agents learn and which proposed lessons they retain.
- Evaluated on AppWorld, tau2-Bench, and Terminal-Bench 2 with component and feedback ablations. On AppWorld (168 tasks; 56 scenarios per run), DuoGate with MiniMax M2.7 increased mean task completion from 60.7% to 69.2% and scenario completion from 41.7% to 54.2% versus ACE over three runs.

Workplace Task Generation for Language-Agent Post-Training

- Developing a GDPval-inspired task corpus for language-agent post-training, using O*NET occupations and public workplace documents to construct realistic requests with supporting files, explicit deliverables, and criterion-level evaluation rubrics.
- Implemented a WorkGenesis-based pipeline for source discovery, supporting-file synthesis, and task authoring, with provenance records, reproducible occupation schedules, and checks linking task requirements to source evidence.
- Built resumable parallel generation with checkpointed outputs, source deduplication, bounded retries, and file-integrity checks. The broader pipeline supports independent solving, rubric-based judging, defect attribution, and teacher-trajectory collection for subsequent training-data curation.

RESEARCH PROJECTS

Pediatric Brain Tumor Classification from Whole-Slide Histopathology

- Developed patient-level models for pediatric brain-tumor classification, aggregating frozen CONCH patch embeddings through gated-attention multiple-instance learning and incorporating age and anatomical location as clinical predictors.
- Curated and audited a 255-patient cohort with 260 usable whole-slide images, reconciling patient identifiers, slide records, and diagnostic labels. Compared hard hierarchical routing, global prediction with coarse-class auxiliary supervision, and strict two-stage training across ten diagnostic classes.
- Conducted fivefold development experiments with fixed patient partitions and internal validation for selected variants. Analyzed class imbalance, per-class errors, and propagation of coarse-class mistakes to characterize the limitations of hierarchical classification.

Geographic Variation in COPD Hospitalization and Outpatient Utilization

- Built a reproducible analysis of Texas inpatient and outpatient encounter data for 44 Austin ZIP codes across 22 quarters (2020 Q1 - 2025 Q2), harmonizing diagnostic definitions, applying official source corrections, and reconciling historical counts.
- Quantified COPD hospitalization relative to observed non-emergency outpatient utilization using observed-to-expected ratios and descriptive Poisson intervals; assessed geographic differences using Fisher exact tests, Holm correction, and quarter-stratified Cochran-Mantel-Haenszel analyses.
- Conducted temporal and case-definition sensitivity analyses and characterized demographic composition and coded discharge mortality. Produced reproducible maps, figures, and manuscript sections to support interpretation of geographic differences in recorded COPD care.

Shared Exploration for Token-Efficient Coding Agents

- Designed a task-group architecture in which similar, complete coding tasks share a persistent exploration agent while retaining separate planning, implementation, and evaluation workflows, with the aim of reducing repeated retrieval and context-processing costs.
- Implemented reproducible task grouping from public requirements using TF-IDF and cosine similarity, with asynchronous evidence exchange and task-scoped access controls. Enabled planners to request additional exploration while preserving each task as an independent unit.
- Built a CodeProjectEval harness for matched comparisons of shared and independent exploration, with isolated outputs, official-test scoring, and request-level token accounting that counts shared exploration once per group; added checks for startup failures and missing usage.

ACADEMIC SERVICE

ICML 2026 - Conference Reviewer May 2026 - Present

- Reviewed machine learning research submissions with attention to originality, technical soundness, experimental design, reproducibility, related work, clarity, and limitations.
- Provided structured feedback on methodology, baselines, evaluation metrics, ablation design, and empirical evidence to support fair and constructive peer review.